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Analyzing Deep Learning in Quick, Draw!: A Case Study on Hand-drawn Sketch Recognition

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Abstract*—This research is motivated by the need to explore alternative models to Convolutional Neural Networks (CNN) in the task of hand-drawn sketch recognition, specifically using the Quick, Draw! dataset. The aim of this research is to evaluate whether a Vision Transformer (ViT) model, ViT-S/16, can match or even outperform the performance of CNN (ResNet-50) in sketch recognition. The research was conducted experimentally by comparing the performance of both models in four different scenarios based on the number of categories used. The results show that ViT is able to approach CNN's performance and even surpass ResNet-50 in one of the experiments. However, as the number of categories increases, the performance of ViT tends to decline and CNN once again shows its superiority, especially in experiments with a larger number of categories. In conclusion, ViT shows potential as an alternative in sketch recognition, but still faces challenges in scenarios with many categories, such as in the Quick, Draw! dataset which has a total of 345 categories. Further adjustments to the architecture or ViT approach are needed to compete more optimally in the future.

***Keywords*—Hand-drawn Sketch Recognition, ResNet-50, ViT-S/16, Quick, Draw!, Image Classification, Model Comparison**

I. INTRODUCTION

The concept of Artificial intelligence has evolved and spans both offline and online games. Recognition models are also widely used in various games, with one popular example being the use of Hand-drawn Sketch Recognition in the game *Quick, Draw!*. Unlike traditional Hand-drawn Sketch Recognition that focus on structured text, Quick, Draw! utilize deep learning techniques, such as Convolutional Neural Networks (CNNs) and other methods to identify and define players' hand-drawn sketches. Deep learning, as a powerful subset of machine learning, encompasses various techniques and architectures, including convolutional neural networks (CNNs) and transformers. Its use has been done for things like handwriting recognition to hand-drawn sketch recognition in the Quick, Draw! game; these deep learning models have completely transformed the field of image recognition [1]. Within transformers, one example is Sketchformer, a transformer-based model that significantly improves sketch recognition performance by modeling the structural and sequential properties of vector sketches more effectively than traditional RNN-based approaches [2].

This study examines how deep learning advances hand-drawn sketch recognition in *Quick, Draw!*, with the goal of identifying an effective deep learning model that can improve the accuracy in predicting and recognizing players' sketches. It explores the misclassification of the model that evaluates to improve the training and modeling optimization. The outcome of this study could enhance the efficiency of hand-drawn sketch recognition in games and various other applications.

A key challenge in recognizing hand-drawn sketches in choosing the most suitable deep learning model to accurately classify sketches from the *Quick, Draw!* dataset. While *Convolutional Neural Networks* (CNN) are a popular choice for image recognition, their effectiveness in handling incomplete and highly varied sketches is still uncertain. Considering the complexity of hand-drawn inputs, exploring alternative models like *Recurrent Neural Networks* (RNN) or combining different approaches could offer better accuracy and efficiency. Previous comparative studies, such as those evaluating CNNs and Vision Transformers in image

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classification and medical imaging tasks [3], [4], [5], highlight the potential of transformer-based models to outperform CNNs in certain complex visual recognition tasks. Additionally, analyzing the exchanges between model performance, computational cost, and real-time processing capabilities is key to determining the most suitable method for large-scale sketch recognition.

An earlier study [6], [7], compared Recurrent Neural Networks (RNN) and Convolutional Neural Networks (CNN) on image classification, which is relevant to our experiment. The study used lots of relevant datasets to image classification, such as the ImageNet dataset and the MINIST dataset. The results of training and testing on those two datasets show that CNN model is better than RNN model in image classification.

This study investigates the use of Vision Transformer (ViT) as a substitute model for hand-drawn sketch recognition, building on existing findings. ViT's accuracy and computing efficiency will be compared against Quick, Draw!'s current CNN model. The goal is to find out if ViT can boost real-time sketch classification in the game and offer performance enhancements.

II. LITERATURE REVIEW

The use of the Vision Transformer model in image classification has been researched, and it has demonstrated that ViT is better than traditional Convolutional Neural Networks (CNNs) when applied to large-scale datasets [8], [9]. This finding is particularly relevant to our research, as the Quick, Draw! game datasets have over 50 million drawings, which is a very large amount. Unfortunately, ViT has high computational costs, which might restrict its wider usage in spite of these benefits. Our study's objective is to apply ViT to the recognition of hand-drawn sketches and evaluate its performance against Quick, Draw!'s default CNN model.

In the study of *A Hybrid Convolutional Neural Network for Sketch Recognition*, Xingyuan Zhang et al. [10] experimented with a Hybrid CNN architecture that combines two networks, A-Net and S-Net. A-Net is used to represent appearance information, and S-Net is used to represent shape information, aiming to improve sketch recognition performance. This approach proves that combining appearance and shape representations can be beneficial for sketch classification and retrieval tasks, as evidenced by experiments on the TU-Berlin, Sketchy, and Flickr15k datasets. However, our research will focus on comparing two different architectural models: Convolutional Neural Networks (CNN), the baseline model in Quick, Draw!, and Vision Transformer (ViT), which is expected to bring a new perspective on sketch understanding through a transformer-based approach. Our study will explore the comparison between CNN and ViT to evaluate both performances in sketch recognition.

Convolutional Neural Networks (CNN) and Vision Transformers (ViTs) have their own advantages and challenges in developing AI-based games with hand-drawn images. CNN is known to be effective in extracting local features through convolution operations, thus being able to recognize detailed and complex visual patterns. This advantage makes CNN suitable for recognizing hand-drawn images that have intricate details. [11] However, CNN has limitations in understanding global relationships in images, which is important for capturing overall patterns and complex shape variations. [12] Traditional CNNs struggle with large-scale, high-resolution images, prompting ongoing research to improve their efficiency and robustness.

In contrast, ViTs uses a self-attention mechanism that allows the model to effectively understand global relationships in images. This method offers increased learning capacity and flexibility in recognizing different image variations. However, ViTs is less than optimal in capturing local correlations that are important in understanding the details of hand-drawn images. To overcome this limitation, a hybrid CNN-Transformer architecture is developed that combines the advantages of CNN in capturing local features and ViTs in understanding global relationships. This approach has shown promising results in improving the accuracy and generalization of the model in hand-drawn image recognition in AI-based games.

Not only applied in gaming technology, Vision Transformer (ViT) has also been explored in histopathology, the gold standard for cancer diagnosis [13]. Due to the unstructured nature of histopathological images, conventional deep learning models often fail to capture fine-grained details. To address this, a recent study proposed Fourier ViT, which combines Fourier Transform and ViT to enhance feature representation and classification accuracy. The model introduces a mixed attention mechanism and outperforms both CNN-based and Transformer-based approaches.

Recently, research on Vision Transformers (ViTs) has grown rapidly due to their success in computer vision tasks [14]. ViTs have shown strong performance across various benchmarks, including tasks like classification, segmentation, and video processing. To enable practical deployment, efficient transformer variants have been proposed, though challenges in scalability and optimization remain.

III. METHODOLOGY

A. Data Collection

The Quick, Draw! dataset is the largest doodle dataset that has been uploaded by Google [15]. The dataset contains 50 million drawings across 345 categories that have been drawn by 15 million people. Every time players finish the game, their drawings will be added to the dataset so that the game dataset keeps increasing, which makes the machine learning model improves time by time. Some representative categories can be seen in Fig. 1.

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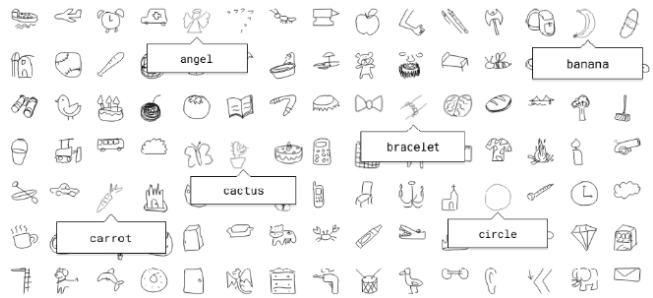


Fig. 1. Several categories in the Quick, Draw! Dataset.

In order to compare the CNN model with ViT using the Quick, Draw! dataset, we conducted all three experiments with 4 different numbers of categories containing 1,000 samples per category. The first experiment involved 10 categories, the second experiment involved 20 categories, and the third experiment involved 30 categories, and the final experiment involved 50 categories. For each experiment, we split 80% for training, 10% for validation, and the rest for testing to ensure a fair evaluation.

B. Architecture Models

Convolutional Neural Network (CNN) in doodle recognition or hand-drawn sketch recognition has been widely used in previous research [16]. Many comparative experiments have been carried out comparing CNN with other models, but so far CNN is still the most efficient in hand-drawn sketch recognition. To find a model that can outperform the CNN model, we compare two architecture models in this experiment: ViT-S/16, a variant of the Vision Transformer model, and ResNet-50, a CNN-based model. Both models have differences in the process of feature extraction and image classification, resulting in interesting new insights into hand-drawn sketch recognition.

1. ViT-S/16 (Vision Transformer model)

[17] Vision Transformer (ViT) begins processing an image by first splitting it into patches of a specified size and converting each patch into a token through linear projection. A special classification token is also appended to the token sequence. Since the self-attention mechanism in transformer encoders lacks inherent position awareness. While positional information is crucial in vision tasks, ViT incorporates positional embeddings into each token. These tokens are then fed through a series of stacked transformer encoder blocks. Each encoder block consists of multi-headed self-attention followed by a feed-forward network.

2. ResNet-50 (CNN model)

[18] The Residual Neural Network (ResNet), which secured the top position in the ImageNet Large Scale Visual Recognition Challenge (ILSVRC 2015), applied the idea of residual connections across its layers. This approach aids in minimizing loss, preserving acquired knowledge, and improving training efficiency. A residual connection refers to the process where the output of a layer is generated by applying convolution to the input and then adding the input back to the result.

C. Experimental Setup

To evaluate the models' performance, we designed an experimental setup that includes data preprocessing and training configuration. This setup ensures consistent input preparation and efficient training. The following subsections describe each component in detail.

1. Image Preprocessing

All images in the Quick, Draw! dataset were resized to 224 × 224 pixels to fit the input requirements of the ViT-S/16 model. This resizing allows the image to be split into 16×16 non-overlapping patches, creating a total of 196 tokens. Additionally, the images were converted to grayscale and normalized to ensure consistent pixel values and stable convergence during training.

2. Model Training Configuration

This study utilized the ViT-S/16 model architecture, initialized with pre-trained weights from ImageNet-21k and subsequently fine-tuned on the Quick, Draw! dataset.

We choose the AdamW optimizer for its ability to enhance generalization by separating weight decay from the gradient update, making it especially well-suited for transformer-based models like ViT-S/16. A learning rate of 1e-5 is used to promote stable convergence, ensuring training stability over time. A batch size of 64 is selected as a practical compromise, providing reliable gradient estimates while ensuring computational efficiency, as indicated by the batch size used in the data loaders.

The training process spans 15-20 epochs, which provides a balanced trade-off between learning efficiency and computational resource usage. This number of epochs is generally sufficient for the model to learn meaningful patterns from the Quick, Draw! dataset without leading to overfitting, especially given the relatively limited complexity of the hand-drawn sketches. For the loss function, cross-entropy loss is employed, a standard and effective choice for multi-class classification problems, making it well-suited for distinguishing between various sketch categories. The implementation is conducted using PyTorch in combination with the timm library, which provides streamlined and optimized support for the ViT-S/16 architecture, enabling efficient experimentation and stable model training.

To enhance generalization and reduce overfitting, data augmentation techniques are applied during preprocessing. Specifically, RandomHorizontalFlip is used to randomly flip images horizontally, allowing the model to learn features that are invariant to orientation changes. Additionally, RandomRotation is applied to

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slightly rotate the images within a fixed degree range (±10°), helping the model become more robust to slight variations in the drawing angle, which is common in hand-drawn sketches. These augmentations introduce variability into the training data, which encourages the model to learn more generalized features.

The chosen setup aligns with established strategies for training Vision Transformers effectively, especially when working with datasets that are smaller in size compared to those at the ImageNet scale, as highlighted in prior research on ViT models [19].

3. Evaluation Metrics

Since this study uses a quantitative research design, numerical metrics that measure prediction accuracy and classification reliability objectively are used to evaluate model performance. Metrics such as accuracy, precision, recall, and F1-score are commonly used in machine learning research to test the model's performance. These metrics provide a clear assessment of the model's ability to make accurate predictions and reduce classification errors, and they also play a significant role as differentiators in the context of hand-drawn sketch detection when comparing ViT with CNN [20].

This experiment is primarily focused on accuracy and loss, even though recent studies [21] highlight the importance of comprehensive metrics like precision, recall, and F1-score, especially in tasks like handwritten text or sketch recognition. The aim is to evaluate general performance differences between ViT and CNN architectures.

a) accuracy

This metric evaluates the percentage of accurate predictions out of all the predictions made:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

As shown in (1), accuracy gives a quick summary of overall performance by measuring the proportion of correct predictions, where TP represents true positives, TN stands for true negatives, FP refers to false positives, and FN indicates false negatives.

b) loss

This metric evaluates how far off the model's predicted probabilities are from the actual labels in classification tasks:

$$Loss = -\sum_{\{i=1)}^{n} y_i \, log(\hat{y}_i) \quad (2)$$

As shown in (2), loss provides detailed feedback during training by quantifying the difference between the true labels □□ and the predicted probabilities $\hat{y}_i$. In classification problems, cross-entropy loss is commonly used because it penalizes incorrect predictions with higher confidence more strongly.

c) parameter size and inference time

Model efficiency was evaluated by its parameter size and average inference time. The total number of learnable parameters was calculated following established model compression methodologies [22], while computational speed was measured using standard inference benchmarking protocols [23].

4. Expected Result

We will investigate the performance of Vision Transformer in sketch recognition, with the expectation that its attention mechanism will enable it to capture complex relationships between image elements more effectively than Convolutional Neural Networks. However, we will also examine whether CNN might still outperform ViT in recognizing simpler or noisier sketch images, where its localized feature extraction could offer an advantage.

IV. RESULT

In this experiment, we tested the models with different amounts of data and a wider range of categories to see how they perform on different types of sketches. The main target was to find out if the models could still work well when faced with more variety, since in real cases like the Quick, Draw! game, the inputs can be really varied and unpredictable. game, there are lots of different things people can draw in many different ways. This is important since the Quick, Draw! game has a lot of categories and many different ways people draw the same thing. So, by trying out different setups, we wanted to see the strengths of each model when facing both more categories and more varied samples. For all of the experiments, we used a learning rate of 10^{-5}.

A. Experiment 1 (10 Categories)

In the first experiment, we started with a standard setup by using 10 categories and 1,000 samples per category and 15 epochs. We chose this approach to see how well each model could perform in simple and balanced conditions. By keeping the number of categories low and the sample size consistent, we could focus on how the models learn patterns without too much complexity. This helped us understand their initial performance before they faced challenging experiments with more categories and fewer samples. The result shows that ViT-S/16 is superior with an accuracy of 94.60% and a loss of 0.1813, while ResNet-50 also gained the same accuracy of 94.60% but with a slightly higher loss of 0.1815. Since the difference is very small, The comparative analysis of 1st experiment can be seen in Fig. 2.

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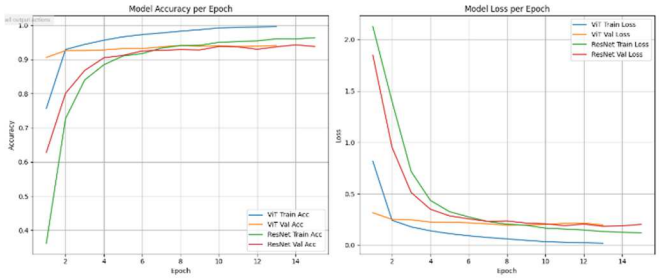


Fig. 2. Accuracy (left) and loss (right) per epoch on training and validation sets for ViT-S/16 and ResNet-50 in the 1st experiment.

B. Experiment 2 (20 Categories)

In the second experiment, we increased the number of categories to 20 to observe the model's learning pattern on more varied data, and we trained the models for 15 epochs. With the results, ViT-S/16 temporarily excels with an accuracy of 94.15% and a loss of 0.2389, while ResNet-50 achieves an accuracy of 93.70% with a loss of 0.2235. The comparative analysis of 2nd experiment can be seen in Fig. 3.

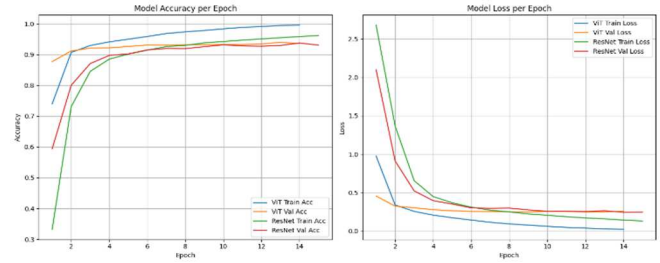


Fig. 3. Accuracy (left) and loss (right) per epoch on training and validation sets for ViT-S/16 and ResNet-50 in the 2nd experiment.

C. Experiment 3 (30 Categories)

In the third experiment, we decided to increase the number of categories to 30 to see which model is superior in handling category variation, and we still trained the models for 15 epochs. The results showed that ResNet-50 was superior in accuracy with 91.10% and a loss of 0.3521, while ViT-S/16 gained an accuracy of 90.37% with a lower loss of 0.3179. The comparative analysis of 3rd experiment can be seen in Fig. 4.

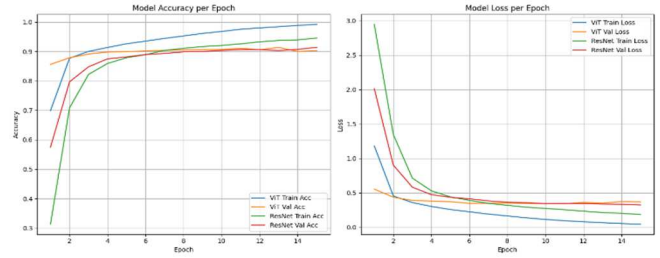


Fig. 4. Accuracy (left) and loss (right) per epoch on training and validation sets for ViT-S/16 and ResNet-50 in the 3rd experiment.

D. Experiment 4 (50 Categories)

In the last experiment, we used 50 categories, but we increased the number of training epochs to 20 to improve training efficiency, considering the larger number of categories. The results showed that ResNet-50 was slightly superior with an accuracy of 88.18% and a loss of 0.4284, while ViT-S/16 obtain accuracy of 88.04% with a loss of 0.4530. This is enough to prove that ResNet-50 continues to perform better when dealing with a larger number of categories, showing more consistent and reliable results as the complexity of the classification task increases. The comparative analysis of 4th experiment can be seen through Fig. 5.

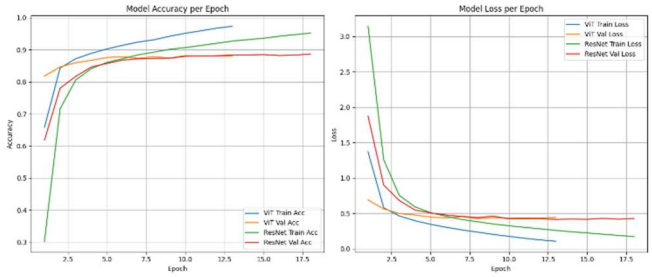


Fig. 5. Accuracy (left) and loss (right) per epoch on training and validation sets for ViT-S/16 and ResNet-50 in the 4th experiment.

To evaluate the classification performance of the models under varying conditions, four experimental scenarios were conducted with different data configurations. The corresponding top-1 validation accuracy and loss values for ViT-S/16 and ResNet-50 across all scenarios are summarized in TABLE I.

TABLE I. VALIDATION ACCURACY (%) AND LOSS OF VIT-S/16 AND RESNET-50 UNDER FOUR DIFFERENT SCENARIOS.

Model	Metrics and Experiment		
	Experiment	Accuracy	Loss
ViT-S/16	1st	94.60	0.1813
	2nd	94.15	0.2389
	3rd	90.37	0.3179
	4th	88.04	0.4530
ResNet-50	1st	94.60	0.1815
	2nd	93.70	0.2235
	3rd	91.10	0.3521
	4th	88.18	0.4284

Based on TABLE I, the decrease in accuracy as the number of classes increases may be due to the higher difficulty in distinguishing between more categories. As the classification task becomes more fine-grained, the chance of misclassification naturally increases. This is especially true when the classes share similar features or patterns. Therefore, more classes lead to a more challenging prediction task, resulting in lower overall accuracy.

E. Perimeter Size and Inference time

Based on the provided metrics, the Vision Transformer model, ViT-S/16, is more efficient in terms of model size, with a smaller parameter count of approximately

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21.7 million compared to ResNet-50's 23.6 million, which suggests that ViT-S/16 is more economical in terms of model parameters. However, this smaller footprint does not translate to faster processing, as its inference time of 0.0032 seconds per sample is slightly slower than ResNet-50's 0.0028 seconds. Although ViT-S/16 has not yet managed to outperform the established ResNet-50 in terms of either inference speed or classification accuracy.

V. CONCLUSION

In this project, we ran several experiments by gradually increasing the number of categories to reflect the real conditions of the Quick, Draw! game, which includes a wide variety of doodles. We kept the number of samples in training consistent and fair across each test. From what we observed, ViT-S/16 gave solid results, but it still couldn't outperform ResNet-50 when it came to recognizing hand-drawn images, especially as the number of categories increased. This performance gap occurred despite ViT-S/16 having a more compact architecture with a smaller parameter count; however, this efficiency in size did not translate to an advantage in speed, as its inference time was slightly slower than ResNet-50's. This shows that ResNet-50 is more reliable for tasks with a lot of variation in the data. Both models have their own strengths: ViT-S/16 works better when there are fewer categories with more samples per class, while ResNet-50 performs more stably when the data involves many different categories. In the future, it would be interesting to try other settings, try mixing both models, or see how they perform on other types of sketch datasets.

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